

Ryan Zeyuan Chen

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📍 Toronto, ON, Canada

Education

Toronto, ON, Canada Sep 2022 – Apr 2025	M.A.Sc. in Mechanical Engineering , University of Toronto Advisor: Prof. Andrew Goldenberg <i>Collaborative Specialization in Robotics</i>
Toronto, ON, Canada Sep 2017 – Apr 2022	B.A.Sc. in Engineering Science , University of Toronto Advisor: Prof. Jessica Burgner-Kahrs <i>Robotics Engineering Major; Artificial Intelligence Engineering Minor; Engineering Business Certificate</i>

Research Experience

Toronto, ON, Canada Sep 2022 – Apr 2025	Robotics Institute, University of Toronto Graduate Researcher <i>Supervised by Prof. Andrew Goldenberg - Robotics and Automation Laboratory</i> - Designed a transfer learning framework to improve data efficiency and generalizability in learning-based kinematics modeling for robots. - Applied the framework to concentric tube continuum robots (CTCRs), reducing the amount of real-world training data needed by 90% for forward kinematics and 33% for inverse kinematics.
Boston, MA, USA May 2023 – Sep 2023	Brigham and Women's Hospital, Harvard Medical School Robotics Research Intern <i>Supervised by Prof. Jayender Jagadeesan - Surgical Imaging and Mechatronics Laboratory</i> - Developed a novel dynamics model for tendon-driven continuum robots (TDCRs) by incorporating static and sliding tendon friction to account for hysteresis effects. - Validated the model through static and dynamic experiments on a physical TDCR prototype. - Integrated a learning-based sub-model into the dynamics pipeline using PyTorch to accelerate computation and improve convergence.
Toronto, ON, Canada May 2021 – Apr 2022	Robotics Institute, University of Toronto Undergraduate Researcher <i>Supervised by Prof. Jessica Burgner-Kahrs - Continuum Robotics Laboratory</i> - Implemented a feedforward neural network (FNN) to learn the forward kinematics of concentric tube continuum robots (CTCRs) using PyTorch, reducing the tip position error by 81% and tip orientation error by 25% compared to state-of-the-art physics-based methods. - Proposed and implemented a learning-based differential inverse kinematics framework for CTCRs, achieving path-following errors of 0.44 mm and 6.21 mm for simple and complex paths in simulation.
Toronto, ON, Canada May 2019 – Apr 2020	Robotics Institute, University of Toronto Undergraduate Researcher <i>Supervised by Prof. Xinyu Liu - Microfluidics and BioMEMS Laboratory</i> - Developed an anti-freezing, ambient-stable, and highly stretchable ionic skin with strong surface adhesion for wearable sensing and soft robotics applications. - Characterized the ionic skin's mechanical, electrical, anti-freezing, and surface adhesion properties through extensive testing under varied environmental conditions. - Demonstrated the piezoresistive sensing capabilities of the ionic skin by integrating it into a wearable glove to measure joint deformation and control a robotic hand via Arduino.

Industry Experience

San Jose, CA, USA Nov 2024 – Present	AMD SerDes Application Design Engineer <i>Technology and Engineering - SerDes Technology - Applications and Validation Team</i> - Resolved firmware bugs and implemented feature enhancements in C++ for existing gigabit transceiver (GT) products based on feedback from customers and internal teams. - Developed firmware unit tests for a new GT, increasing code coverage from 4.4% to 93.7%. - Maintained Vivado design software codebase to support existing GTs, implementing new features and updates in response to firmware changes.
San Jose, CA, USA Jun 2020 – May 2021	Xilinx SerDes Application Design Intern <i>Wired and Wireless Group - Wired System Engineering - Applications and Validation Team</i> - Developed 50+ unit test cases to validate API functionality for FPGA control using C++. - Verified tests on virtual machines and hardware prototypes across multiple communication protocols. - Performed regression testing following hardware upgrades to ensure consistent software performance.

Publications

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| 2022 | 1. Grassmann, R. M., Chen, R. Z. , Liang, N. & Burgner-Kahrs, J. <i>A Dataset and Benchmark for Learning the Kinematics of Concentric Tube Continuum Robots</i> in <i>IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i> (2022), 9550–9557. |
| 2021 | 2. Ying, B., Chen, R. Z. , Zuo, R., Li, J. & Liu, X. An Anti-Freezing, Ambient-Stable and Highly Stretchable Ionic Skin with Strong Surface Adhesion for Wearable Sensing and Soft Robotics. <i>Advanced Functional Materials</i> 31 , 2104665 (2021). |

Presentations

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| 2023 | National Center for Image-Guided Therapy (NCIGT) Summer Research Retreat , Brigham and Women's Hospital
Speaker at the Continuum Robotics session, "Robotic Assisted Intracardiac Echocardiography (ICE) Catheter." |
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Honors and Awards

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| 2021 | Faculty of Applied Science and Engineering Dean's Innovation Fellowship , University of Toronto
Awarded to undergraduate engineering students conducting research in the areas of "Smart Cities, Smart Health, and The Internet of Things." |
| 2019 | Undergraduate Student Research Award , Natural Sciences and Engineering Research Council of Canada (NSERC)
Awarded to undergraduate students pursuing research in the natural sciences and engineering, based on outstanding academic record and demonstrated research aptitude. |
| 2017 | Albert and Rose Jong Entrance Scholarship , University of Toronto
Presented for demonstrating leadership in the Chinese-Canadian community as a first-year student in the Electrical Engineering or Engineering Science program. |
| 2017 | Faculty of Applied Science and Engineering Admission Scholarship , University of Toronto
Awarded for outstanding academic achievement in the prerequisite courses as a first-year engineering student. |

Technical Skills

Robotics Engineering: Robot Modeling, Machine Learning, Control Theory, Computer Vision, Human-Robot Interaction

Software Development: Python, C, C++ , MATLAB, HTML

Tools and Packages: Git, NumPy, PyTorch, Scikit-learn, Pandas, Matplotlib, Seaborn, CMake